

Muhammad Ibrahim

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EDUCATION

New York University <i>Bachelor of Science in General Engineering, Mechanical and Civil Focus</i> - Major GPA - 3.60 / 4.00, UK (2:1). - Major Coursework - Solid Mechanics, Engineering Dynamics, Fluid Mechanics, Heat Transport, Composite Materials.	Abu Dhabi, UAE May 2026
New York University <i>Study Away, Immigration and Theatre</i> Analysed migration in French theatre casting and the economics of integration; Term GPA - 3.70 / 4.00.	Paris, France Jan 2024
New York University June Term <i>Study Away, Architecture and Climate Impact</i> Examined how architectural decisions shape climate outcomes and purpose-led design; Term GPA - 4.00 / 4.00.	Valletta, Malta Jun 2023

EXPERIENCE

Center for Stability, Instability and Turbulence (SITE) <i>Graduate Intern, New York University</i> - Ran a 2D free surface study solo, 60 trials across 4 angles of attack, to build the dataset the model was trained on. - Grounded the predictive model in a dimensional analysis of surface height against velocity, depth and body radius. - Trained a machine learning model in Python and MATLAB predicting free surface profiles beyond the tested range.	New York, USA May 2026 - Jul 2026
Advanced Manufacturing Workshop <i>Workshop Technician, New York University</i> - Cut print failure rates from 50% to 10% across over 70 print hours by diagnosing root causes and refining parameters. - Manufactured polymer and composite parts, reinforcing them with bidirectional carbon fibre, fibreglass and chopped strand.	Abu Dhabi, UAE Sep 2025 - May 2026
Center for Stability, Instability and Turbulence (SITE) <i>Undergraduate Research Assistant, New York University</i> - Tested over 130 fluid instability cases in a 2-person team, probing for linearity between laminar and turbulent regimes. - Developed water-tunnel models that reliably triggered boundary layer separation, quantified by flow visualisation.	Abu Dhabi, UAE Apr 2023 - May 2026
Energy and Propulsion Lab <i>Undergraduate Research Assistant, New York University</i> - Solved Computational Fluid Dynamics problems in Converge and OpenFOAM on HPC clusters. - Simulated airflow through vertical growing towers for AeroFarms, returning a 28% ventilation gain at upper levels. - Presented Urban Air Mobility propulsion conclusions to a 25-member Hyundai NGV panel in South Korea.	Abu Dhabi, UAE Nov 2022 - Nov 2023

PROJECTS

Fixed-Wing UAV for Search and Rescue Payload Deployment , <i>Personal Project</i> - Owned the airframe, structure and materials on a 1130 mm wingspan 3D printed UAV, built with one collaborator. - Reduced airframe mass from 2500 to 1500 g via lightweight PLA, infill tuning, internal geometry and carbon tubing. - Designed a release mechanism to drop a multirotor payload in flight, delivered as a completed prototype.	<i>Jun 2026 - Present</i>
Advanced Anti-Roll Tank System (ART3S) , <i>Mubadala Investment and Abu Dhabi Maritime Academy</i> - Engineered SolidWorks models of 200 to 800 cubic metre anti-heeling tanks for full-scale ferry and car-carrier vessels. - Modelled vessel roll decay in Simulink, cutting decay time 33% with passive tanks and 62% with active control.	<i>Oct 2024 - Sep 2025</i>
Savonius Turbine for Traffic Wind Energy Harvesting , <i>Capstone Design</i> - Optimised a vertical-axis wind turbine in ANSYS Fluent with a 45-degree helical twist and serrated leading edges. - Raised energy recovery from 56% of the theoretical maximum to 80% by adding the serrations. - Measured 3 to 15 W output at 1 to 5 m/s wind from a 1:3 scale carbon fibre FDM prototype.	<i>Sep 2024 - May 2025</i>
Compressor-Less Multi-Stage TEC Refrigerator , <i>New York University</i> - Replaced compressor and refrigerant with 5 stacked thermoelectric modules, cooling 27 to 10 degrees in 24 minutes. - Tuned a closed-loop PID controller on Arduino, holding the 10 degree setpoint within the ±0.5 degree sensor accuracy.	<i>Mar 2025 - May 2025</i>
Formula Student SAE Drivetrain Design , <i>New York University</i> - Exposed a half-shaft 49% over its twist limit before build, by simulating shock loads that static checks had missed. - Redirected the redesign to geometry, saving the cost of an alloy with zero twist gain, by proving stiffness ignores yield.	<i>Sep 2024 - May 2025</i>

AWARDS

Research Grant , <i>Mubadala and Abu Dhabi Maritime Association</i>	<i>Aug 2024</i>
Industry Recognition and Grant , <i>Hyundai NGV</i>	<i>Oct 2022</i>
Full Merit Scholarship , <i>New York University</i>	<i>Dec 2021</i>

EXTRACURRICULAR ACTIVITIES

NYU Abu Dhabi Athletics , <i>Varsity Jiu-Jitsu, Recreational Sport and Team Videographer</i> - Competed in jiu-jitsu at university varsity level, earning selection in each of four years. - Filmed and photographed the university cricket team across a full season, delivering footage and stills each week. - Recreational sport across padel, golf, tennis, cricket, volleyball, badminton, football, cycling and wall climbing.	<i>2022 - 2026</i>
Habitat for Humanity Nepal , <i>Volunteer, Residential Build Programme</i> - Constructed a single-storey house from foundation to roof with a volunteer team, handing a finished home to its family. - Checked dimensions and layouts against the civil drawings on site before the team began work. - Converted donated labour and materials into permanent housing for families priced out of building.	<i>2025</i>
Emirates Motorsports Organization , <i>Chief Track Marshal</i> Directed trackside safety and first response across 15 events at Yas Marina Circuit and Dubai Autodrome.	<i>2019 - 2025</i>

SKILLS

CAD and Design - SolidWorks (CSWA), mechanism design, design for manufacturing, rapid prototyping.
Simulation and Analysis - CFD, finite element analysis (FEA), ANSYS Fluent, OpenFOAM, Converge, Simulink, HPC.
Manufacturing - FDM 3D printing, carbon fibre and fibreglass composites, Arduino, closed-loop PID control.
Programming - Python, MATLAB, machine learning, sensor data acquisition.
Languages - English C2, Urdu C2, Arabic A2.